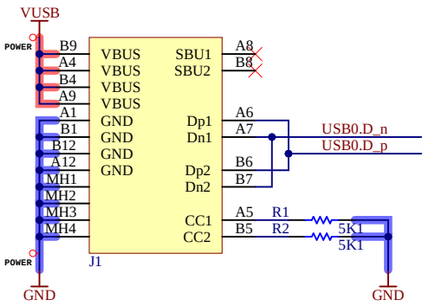
Rev. 00 : Initial release

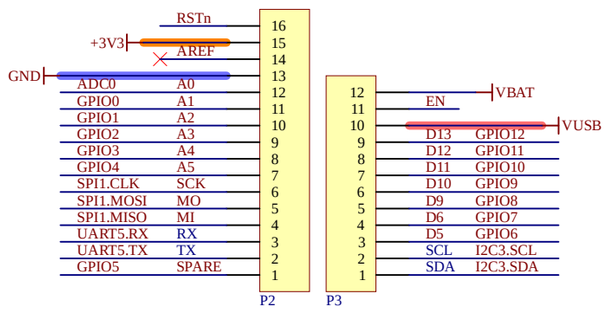
Tess

Allwinner T113-S3 based Feather compatible

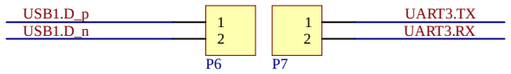
USB-C Interface

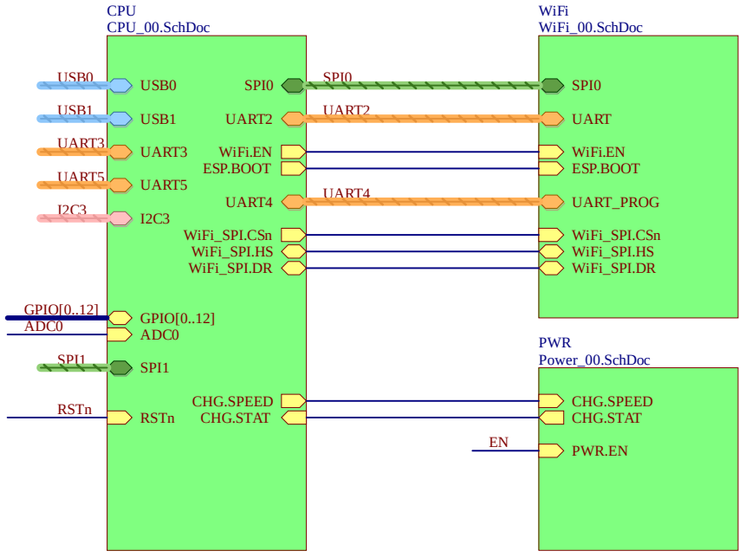


Adafruit Feather



Extra features





Phyx	Title: Tess	
	Date: 28/11/2023	Size: A3
Revision: 00	Dranw By: Wim Van Gool	Sheet 1 of 4

Tess - CPU

T113-S3 Dual core Cortex A7 CPU

USB0 USB0 USB D_p D_n USB0.D_p USB0.D_n
USB1 USB1 USB D_p D_n USB1.D_p USB1.D_n
I2C3 I2C3 I2C SCL SDA I2C3.SCL I2C3.SDA
UART2 UART2 UART4 TX RX RTS CTS UART2.TX UART2.RX UART2.RTS UART2.CTS
UART5 UART5 UART TX RX UART5.TX UART5.RX
UART4 UART4 UART TX RX UART4.TX UART4.RX
UART3 UART3 UART TX RX UART3.TX UART3.RX
ESP.BOOT WiFi.EN CHG.SPEED CHG.STAT ADC0 GPIO[0..12] RSTn
SPI1 SPI1 SPI CLK MISO MOSI SPI1.CLK SPI1.MISO SPI1.MOSI
SPI0 SPI0 SPI CLK MISO MOSI SPI0.CLK SPI0.MISO SPI0.MOSI
WiFi_SPI.CSn WiFi_SPI.HS WiFi_SPI.DR

BOOT_SEL[1:0]
00 : SPI NOR > SPI NAND
01 : SD0 > SPI NOR > Other
10 : SD0 > SPI NAND > Other
11 : SD0 > eMMC2 > eMMC_USR > Other
-> Any option except for 00 is ok!

+3V3 R4 4K7 R5 4K7
BOOT SEL0
BOOT SEL1

GPIO1	86	PB2	LCD0-D0/I2S2-DOUT2/TWI0-SDA/I2S2-DIN2/LCD0-D18/UART4-TX/CAN0_TX0	
GPIO2	85	PB3	LCD0-D1/I2S2-DOUT1/TWI0-SCK/I2S2-DIN0/LCD0-D19/UART4-RX/CAN0_RX0	
UART5.TX	84	PB4	LCD0-D8/I2S2-DOUT0/TWI1-SCK/I2S2-DIN1/LCD0-D20/UART5-TX/CAN1_TX0	
UART5.RX	82	PB5	LCD0-D9/I2S2-BCLK/TWI1-SDA/PWM0/LCD0-D21/UART5-RX/CAN1_RX0	
UART3.TX	80	PB6	LCD0-D16/I2S2-LRCK/TWI3-SCK/PWM1/LCD0-D22/UART3-TX/CPUBIST0	
UART3.RX	79	PB7	LCD0-D17/I2S2-MCLK/TWI3-SDA/IR-RX/LCD0-D23/UART3-RX/CPUBIST1	
SPI0.CLK	19	PC2	SPI0-CLK/SDC2-CLK	
WiFi_SPI.CSn	18	PC3	SPI0-CS0/SDC2-CMD	
SPI0.MOSI	17	PC4	SPI0-MOSI/SDC2-D2/BOOT-SEL0	
SPI0.MISO	16	PC5	SPI0-MISO/SDC2-D1/BOOT-SEL1	
WiFi_SPI.HS	15	PC6	SPI0-WP/SDC2-D0/UART3-TX/TWI3-SCK/DBG-CLK	
GPIO8	14	PC7	SPI0-HOLD/SDC2-D3/TWI3-SDA/TCON-TRIG	
100R DSI.D0_P	55	PD0	LCD0-D2/LVDS0-V0P/DSI-D0P/TWI0-SCK	
100R DSI.D0_N	56	PD1	LCD0-D3/LVDS0-V0N/DSI-D0N/UART2-TX	
100R DSI.D1_P	57	PD2	LCD0-D4/LVDS0-V1P/DSI-D1P/UART2-RX	
100R DSI.D1_N	58	PD3	LCD0-D5/LVDS0-V1N/DSI-D1N/UART2-RTS	
100R DSI.C_P	59	PD4	LCD0-D6/LVDS0-V2P/DSI-CKP/UART2-CTS	
100R DSI.C_N	60	PD5	LCD0-D7/LVDS0-V2N/DSI-CKN/UART5-TX	
100R DSI.D2_P	61	PD6	LCD0-D10/LVDS0-CKP/DSI-D2P/UART5-RX	
100R DSI.D2_N	62	PD7	LCD0-D11/LVDS0-CKN/DSI-D2N/UART4-TX	
100R DSI.D3_P	63	PD8	LCD0-D12/LVDS0-V3P/DSI-D3P/UART4-RX	
100R DSI.D3_N	64	PD9	LCD0-D13/LVDS0-V3N/DSI-D3N/PWM6	
GPIO5	67	PD10	LCD0-D14/LVDS1-V0P/SPI1-CS/DBI-CSX/UART3-TX	
SPI1.CLK	68	PD11	LCD0-D15/LVDS1-V0N/SPI1-CLK/DBI-SCLK/UART3-RX	
SPI1.MOSI	70	PD12	LCD0-D18/LVDS1-V1P/SPI1-MOSI/DBI-SDO/TWI0-SDA	
SPI1.MISO	69	PD13	LCD0-D19/LVDS1-V1N/SPI1-MISO/DBI-SDI/DBI-TE/DBI-DCX/UART3-RTS	
DSI.GPIO0	71	PD14	LCD0-D20/LVDS1-V2P/SPI1-HOLD/DBI-DCX/DBI-WRX/UART3-CTS	
DSI.GPIO1	72	PD15	LCD0-D21/LVDS1-V2N/SPI1-WP/DBI-TE/IR-RX	
GPIO13	73	PD16	LCD0-D22/LVDS1-CKP/DMIC-DATA3/PWM0	
GPIO14	74	PD17	LCD0-D23/LVDS1-CKN/DMIC-DATA2/PWM1	
GPIO4	75	PD18	LCD0-CLK/LVDS1-V3P/DMIC-DATA1/PWM2	
GPIO3	76	PD19	LCD0-DE/LVDS1-V3N/DMIC-DATA0/PWM3	
I2C2.SCL	54	PD20	LCD0-HSYNC/TWI2-SCK/DMIC-CLK/PWM4	
I2C2.SDA	53	PD21	LCD0-VSYNC/TWI2-SDA/UART1-TX/PWM5	
WiFi_SPI.DR	52	PD22	OWA-OUT/IR-RX/UART1-RX/PWM7	

U1D

+3V3 R38 4K7 R39 4K7

X32K Cshunt = 1.1pF
DX Cshunt = 6.5pF

	PE0	44	UART2.RTS
	PE1	45	UART2.CTS
	PE2	35	UART2.TX
	PE3	33	UART2.RX
	PE4	43	UART4.TX
	PE5	42	UART4.RX
	PE6	41	I2C3.SCL
	PE7	40	I2C3.SDA
	PE8	39	USER.LED2
	PE9	38	USER.LED1
	PE10	37	ESP.BOOT
	PE11	36	WiFi.EN
	PE12	32	GPIO6
	PE13	31	GPIO7
SDC0.D1	PF0	7	
SDC0.D0	PF1	8	
SDC0.CLK	PF2	9	
SDC0.CMD	PF3	10	
SDC0.D3	PF4	11	
SDC0.D2	PF5	12	
GPIO9	PF6	13	
SDC1.CLK	PG0	120	
SDC1.CMD	PG1	118	
SDC1.D0	PG2	119	
SDC1.D1	PG3	121	GPIO0
SDC1.D2	PG4	123	
SDC1.D3	PG5	122	
	PG6	1	
GPIO12	PG7	2	
	PG8	3	
GPIO11	PG9	4	
	PG10	5	
GPIO10	PG11	6	
CHG.STAT	PG12	124	
CHG.SPEED	PG13	125	
USER.LED0	PG14	126	
	PG15	127	

U1E

+3V3 R6 4K7 R7 4K7 R11 4K7 R12 4K7 R35 4K7

DSI Connector

DSI.D0_N DSI.D0_P DSI.D1_N DSI.D1_P DSI.C_N DSI.C_P DSI.D2_N DSI.D2_P DSI.D3_N DSI.D3_P DSI.GPIO0 DSI.GPIO1 I2C2.SCL I2C2.SDA +3V3 GND
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22
P4

User LEDs

+3V3 +3V3 +3V3
R23 4K7 R8 4K7 R9 4K7
D1 D3 D4
USER.LED0 USER.LED1 USER.LED2

Microphone

+1V8 2 VDD OUT 1 MIC_IN3_p
GND GND 3 4
U6

Expansion header

+3V3
UART3.TX UART3.RX HP_OUT_L HP_OUT_R
1 3 5 7 9 2 4 6 8 10
GPIO14 GPIO13 I2C2.SCL I2C2.SDA
P5

USB1.D_p 112 USB1-DP
USB1.D_n 113 USB1-DM
USB0.D_p 115 USB0-DP
USB0.D_n 114 USB0-DM
U1F

GND R31 75R TV_OUT0 78 TVOUT0
TV_IN_VR_p 110 TVIN-VRP
TV_IN_VR_n 111 TVIN-VRN
TV_IN1 109 TVIN1
TV_IN0 108 TVIN0
TP_Y2 105 TP-Y2
TP_Y1 104 TP-Y1
TP_X2 103 TP-X2
TP_X1 102 TP-X1
ADC0 101 GPADC0
MIC_IN3_p 87 MICIN3P
MIC_IN3_n 88 MICIN3N
LINE_IN_R 95 LINEINR
LINE_IN_L 96 LINEINL
FM_IN_R 93 FMINR
FM_IN_L 94 FMINL
HP_OUT_R 98 HPOUTR
HP_OUT_L 99 HPOUTL
HP_OUT_FB 100 HPOUTFB
VRA2 90 VRA2
VRA1 92 VRA1
U1C

C59 100nF 50V C60 100nF 50V
C51 100nF 50V C52 100nF 50V C53 100nF 50V C48 100nF 50V C49 100nF 50V
R32 33R R34 33R

X32KOUT 24 32k_OUT
X32KIN 25 32k_IN
DXOUT 22 DX_OUT
DXIN 23 DX_IN
REFCLK-OUT 21
RESET 27 RSTn
NC0 106
DZQ 47
U1B

R33 4K7 +3V3
R30 240R C50 100nF 50V
Y2 C1 10pF C2 10pF Y1 C3 10pF C4 10pF
SW1
Place close to SoC

CAPS

Pull CLK low will disable the storage and enable USB bootloader

SW2 R40 240R

SD-Card

+3V3
TP3 SDC0.D0 6
TP4 SDC0.D1 7
TP5 SDC0.D2 2
TP6 SDC0.D3 1
TP7 SDC0.CMD 5
TP8 SDC0.CLK 3
TP9
TP10
D0/N.C/DO D1/N.C/N.C. D2/N.C/N.C. D3/N.C./CS CMD/CMD/DI CLK/CLK/CLK
4b/1b/SPI
VDD 8 GND 4
U2
+3V3 C5 22uF 6V3 GND

+0V9 C6 100nF 50V C7 100nF 50V C8 100nF 50V C9 100nF 50V C10 10uF 6V3
+1V5 C20 100nF 50V VCC-DRAM C21 1uF 10V VCC-DRAM C22 10uF 6V3 VCC-DRAM C23 2u2F 6V3 LDOB-OUT

+0V9 POWER 117 VDD-CORE1 16 VDD-CORE0 81 VDD-SYS2 51 VDD-SYS1 46 VDD-SYS0 0.9V
+1V5 POWER 30 LDOB-OUT 49 VCC-DRAM1 48 VCC-DRAM0 1.5V
91 AGND 29 EPAD - VSS
U1A

1.8V LDOA-OUT 28 AVCC 89 HPVCC 97 VCC-RTC 26 VCC-PLL 20 VCC-TVIN 107 VDD18-DRAM 50 VCC-LVDS 65 POWER +1V8
3.3V LDO-IN 29 VCC-TVOUT 77
1.8/2.5/3.3V VCC-PE 34 VCC-PD 66 VCC-PG 128 VCC-IO 83 +3V3

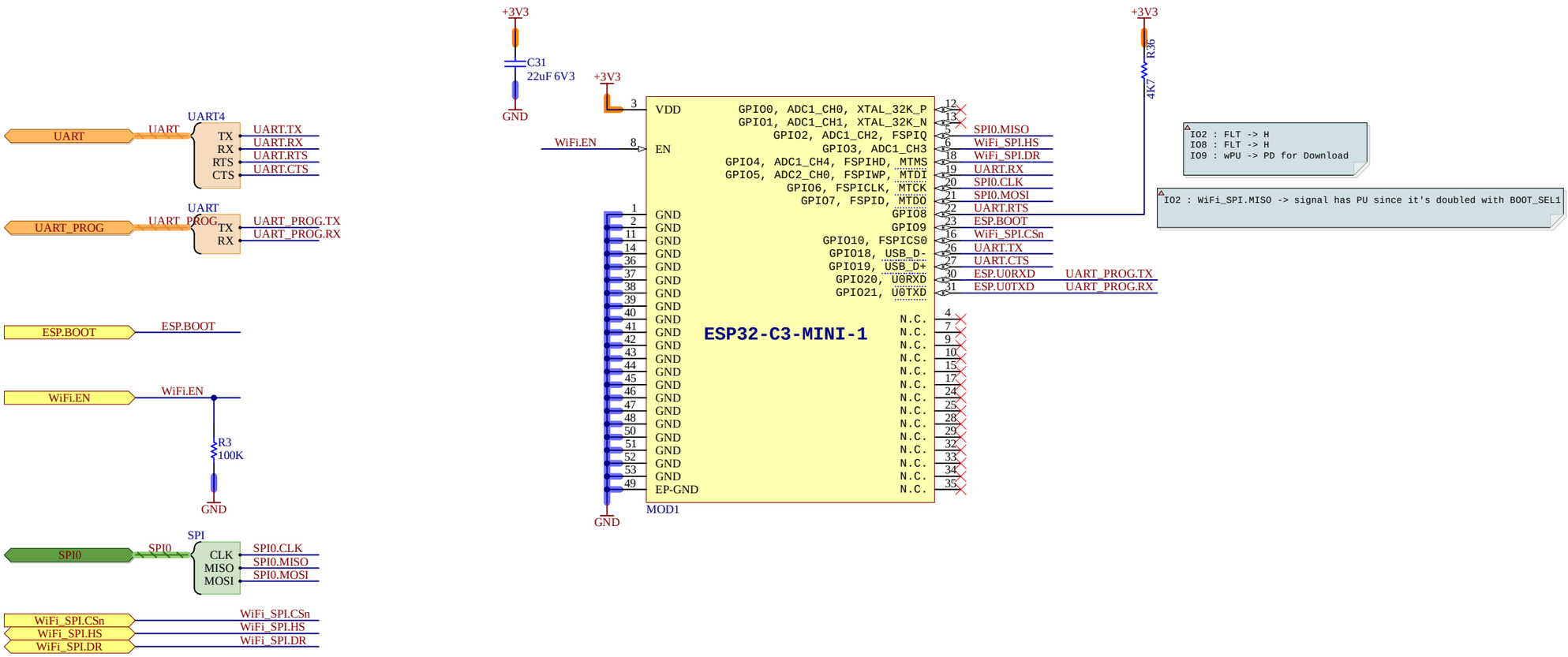
+3V3 C24 2u2F 6V3 LDO-IN C25 100nF 50V VCC-PE C26 100nF 50V VCC-PD C27 100nF 50V VCC-PG C28 100nF 50V VCC-IO C29 100nF 50V VCC-TVOUT C30 10uF 6V3

+1V8 C11 2u2F 6V3 LDOA-OUT C12 100nF 50V LDOA-OUT C13 100nF 50V VCC-PLL C14 2u2F 6V3 AVCC C15 2u2F 6V3 HPVCC C16 100nF 50V HPVCC C17 100nF 50V VCC-TVIN C18 100nF 50V VCC-LVDS C19 100nF 50V VDD18-DRAM

Phyx	Title	Tess - CPU	
	Date	28/11/2023	Size A3
Revision 00	Dranw By	Wim Van Gool	Sheet 2 of 4

Tess - Wi-Fi

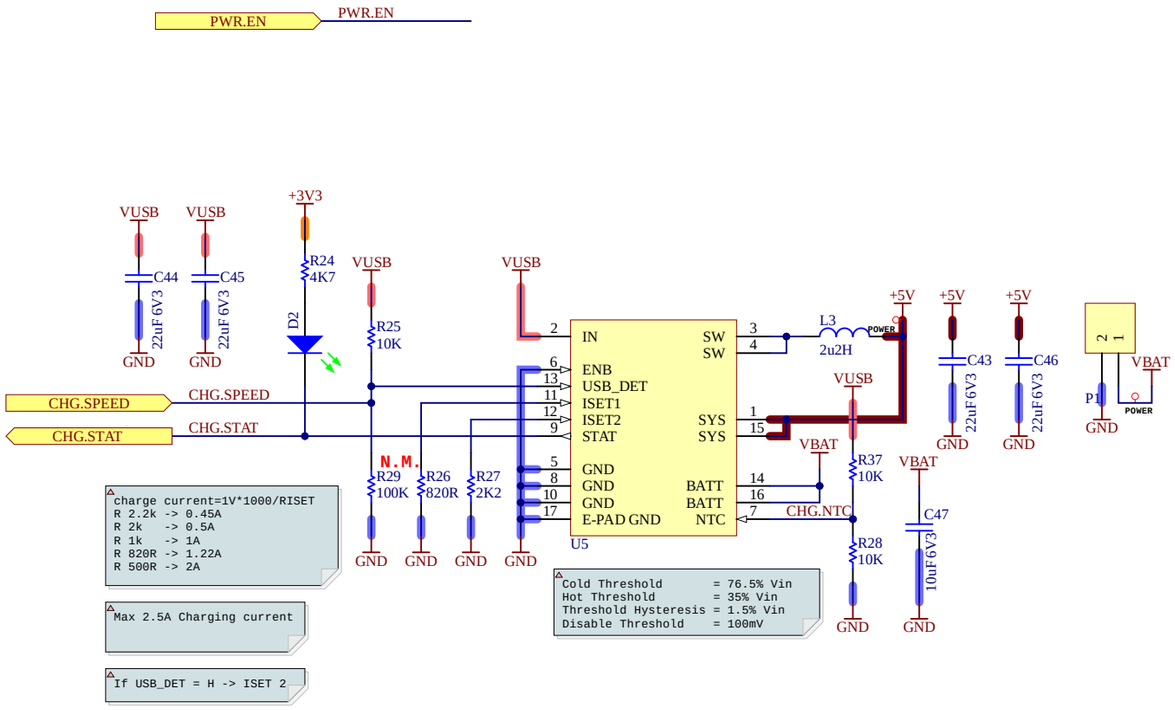
ESP32-C3 Wi-Fi Module

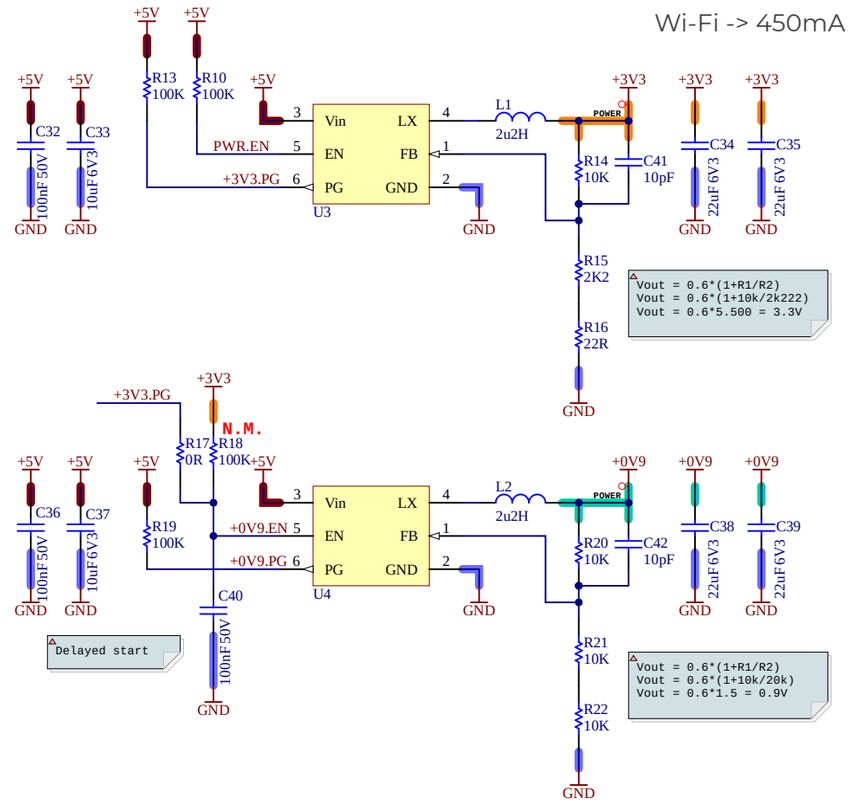


Phyx	Title	Tess - Wi-Fi	
	Date	28/11/2023	Size A3
Revision 00	Dranw By	Wim Van Gool	Sheet 3 of 4

Tess - Power

Power Supplies





Phyx	Title: Tess - Power	
	Date: 28/11/2023	Size: A3
Revision: 00	Dranw By: Wim Van Gool	Sheet: 4 of 4